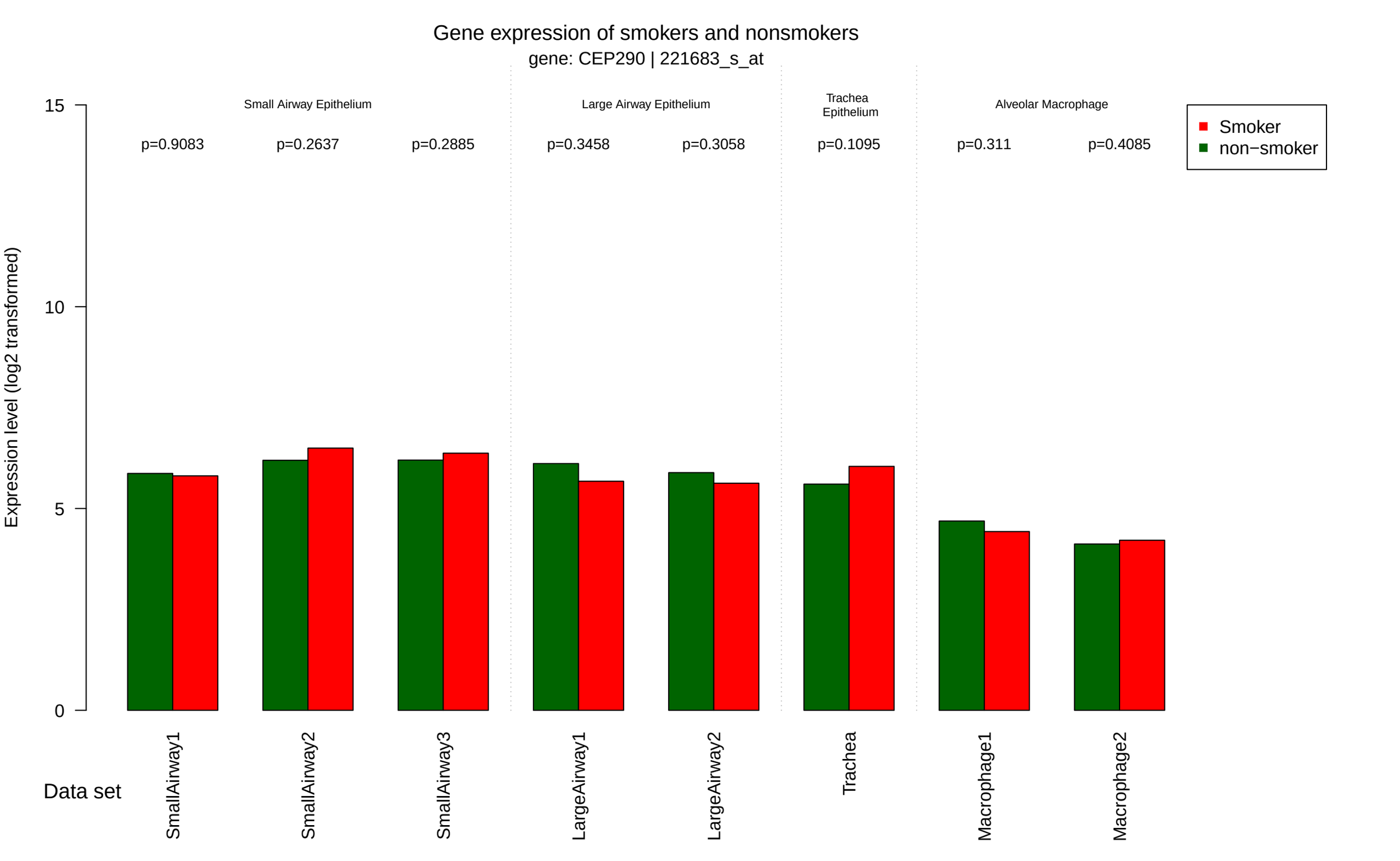
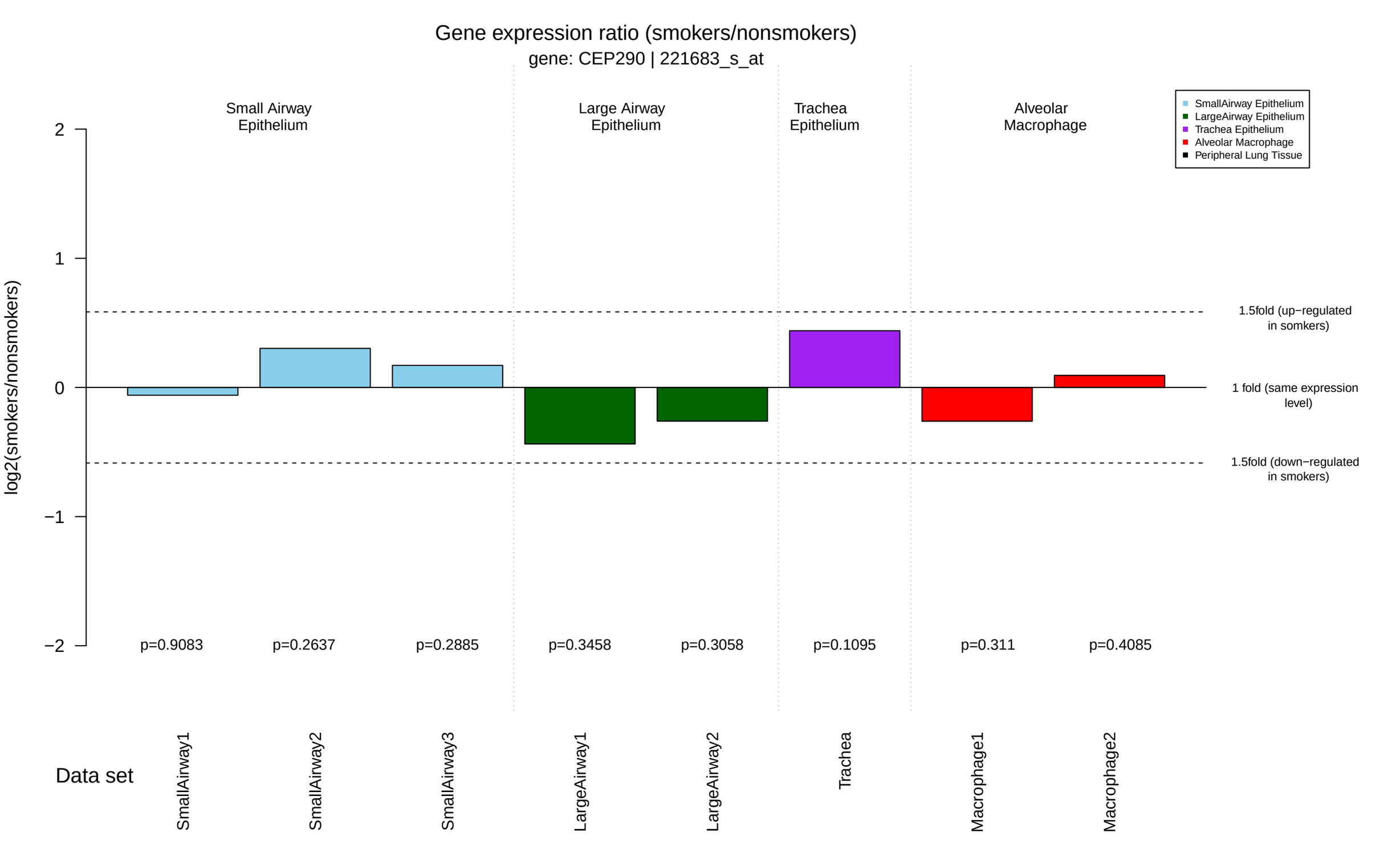
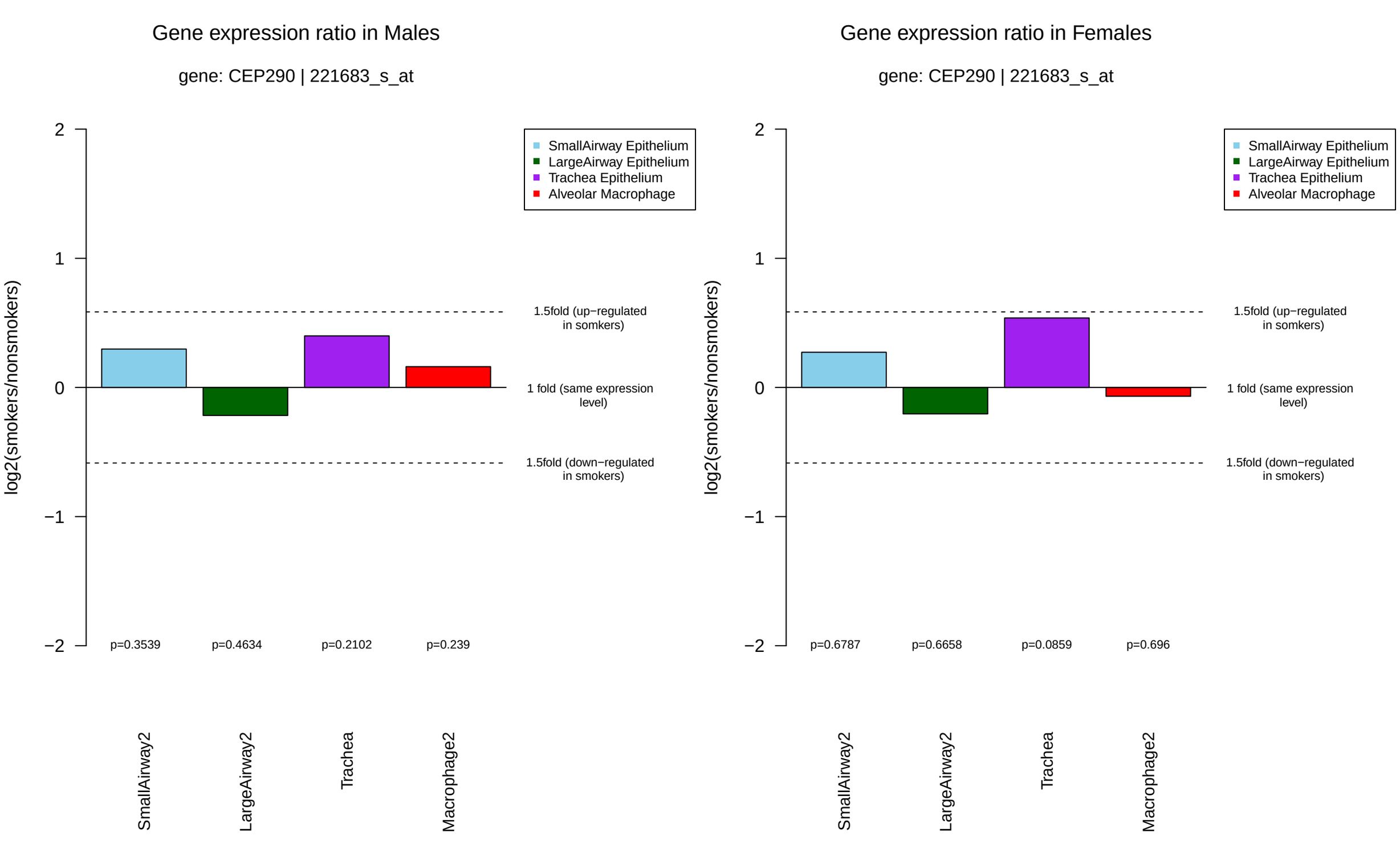
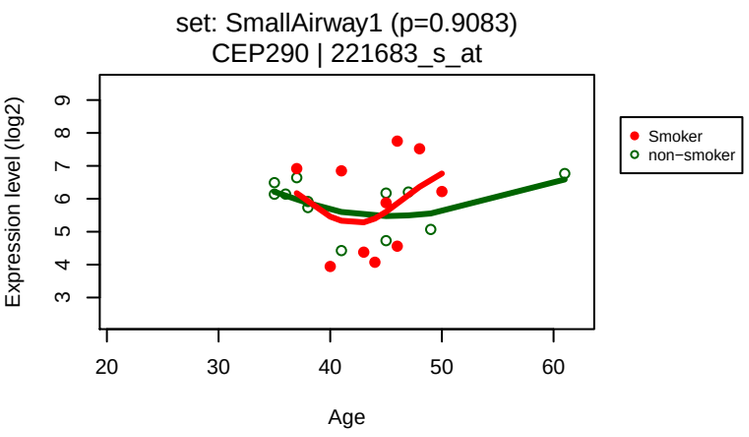
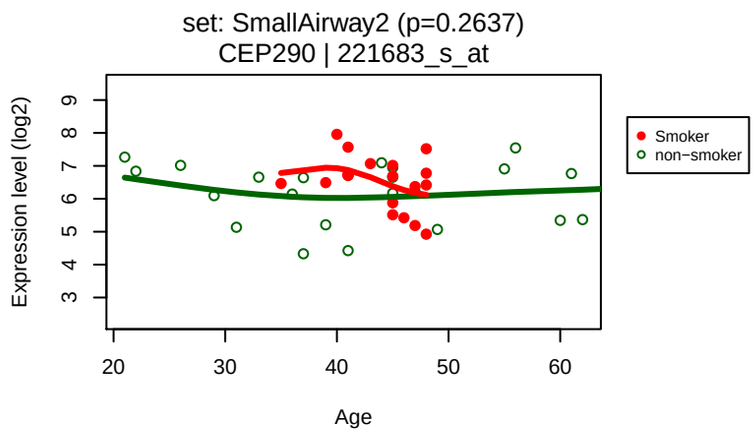
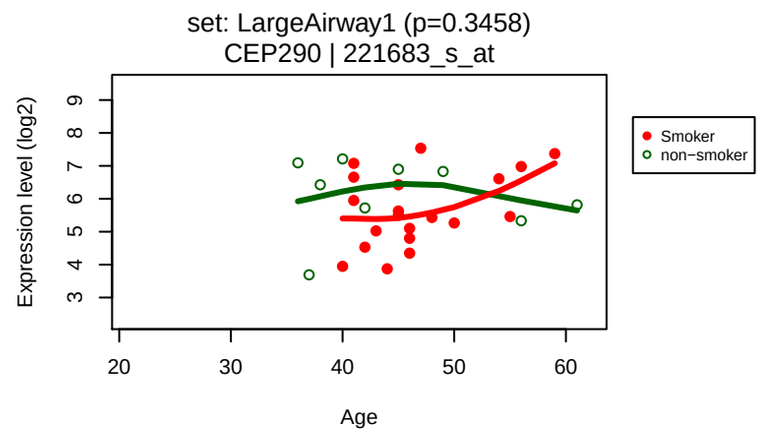

Gene expression of smokers and nonsmokers
gene: CEP290 | 221683_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9083
p=0.2637
p=0.2885
p=0.3458
p=0.3058
p=0.1095
p=0.311
p=0.4085
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

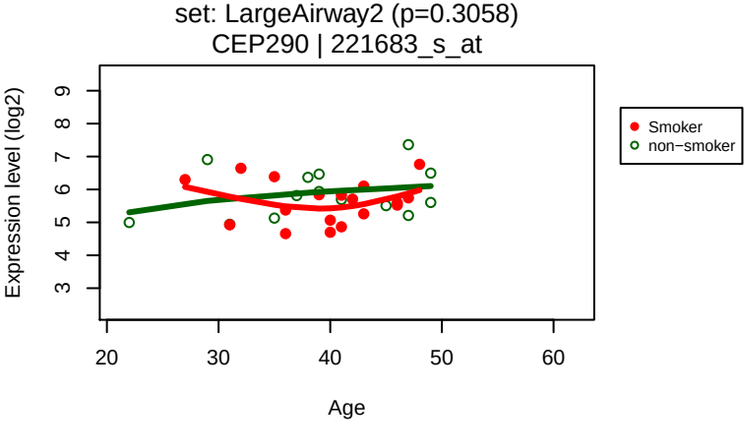

Gene expression ratio (smokers/nonsmokers)
gene: CEP290 | 221683_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9083
p=0.2637
p=0.2885
p=0.3458
p=0.3058
p=0.1095
p=0.311
p=0.4085
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

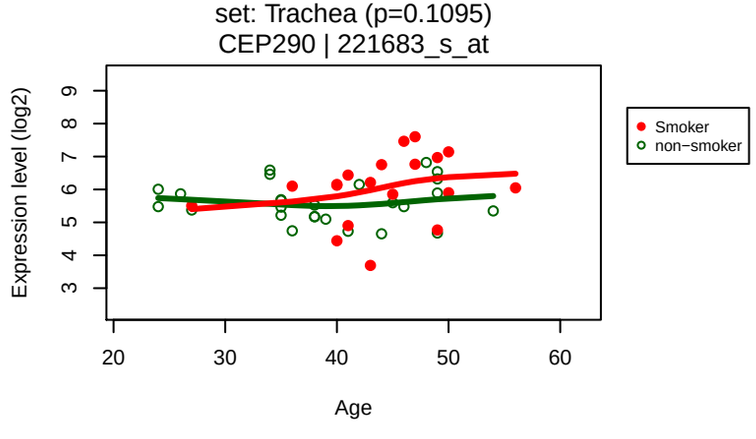

Gene expression ratio in Males
gene: CEP290 | 221683_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3539
p=0.4634
p=0.2102
p=0.239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEP290 | 221683_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6787
p=0.6658
p=0.0859
p=0.696
SmallAirway2
LargeAirway2
Trachea
Macrophage2

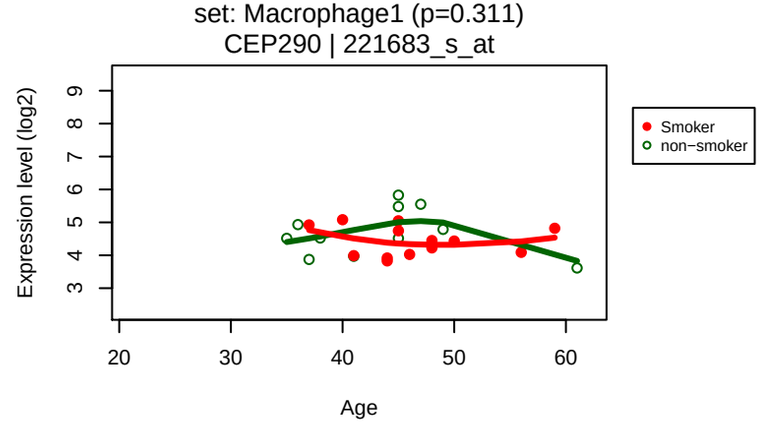

set: SmallAirway1 (p=0.9083)
CEP290 | 221683_s_at
Expression level (log2)
Age
Smoker
non−smoker

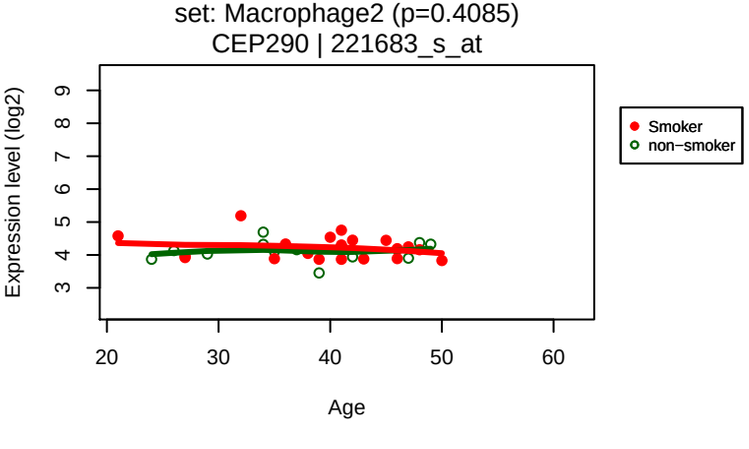

set: SmallAirway2 (p=0.2637)
CEP290 | 221683_s_at
Expression level (log2)
Age
Smoker
non−smoker

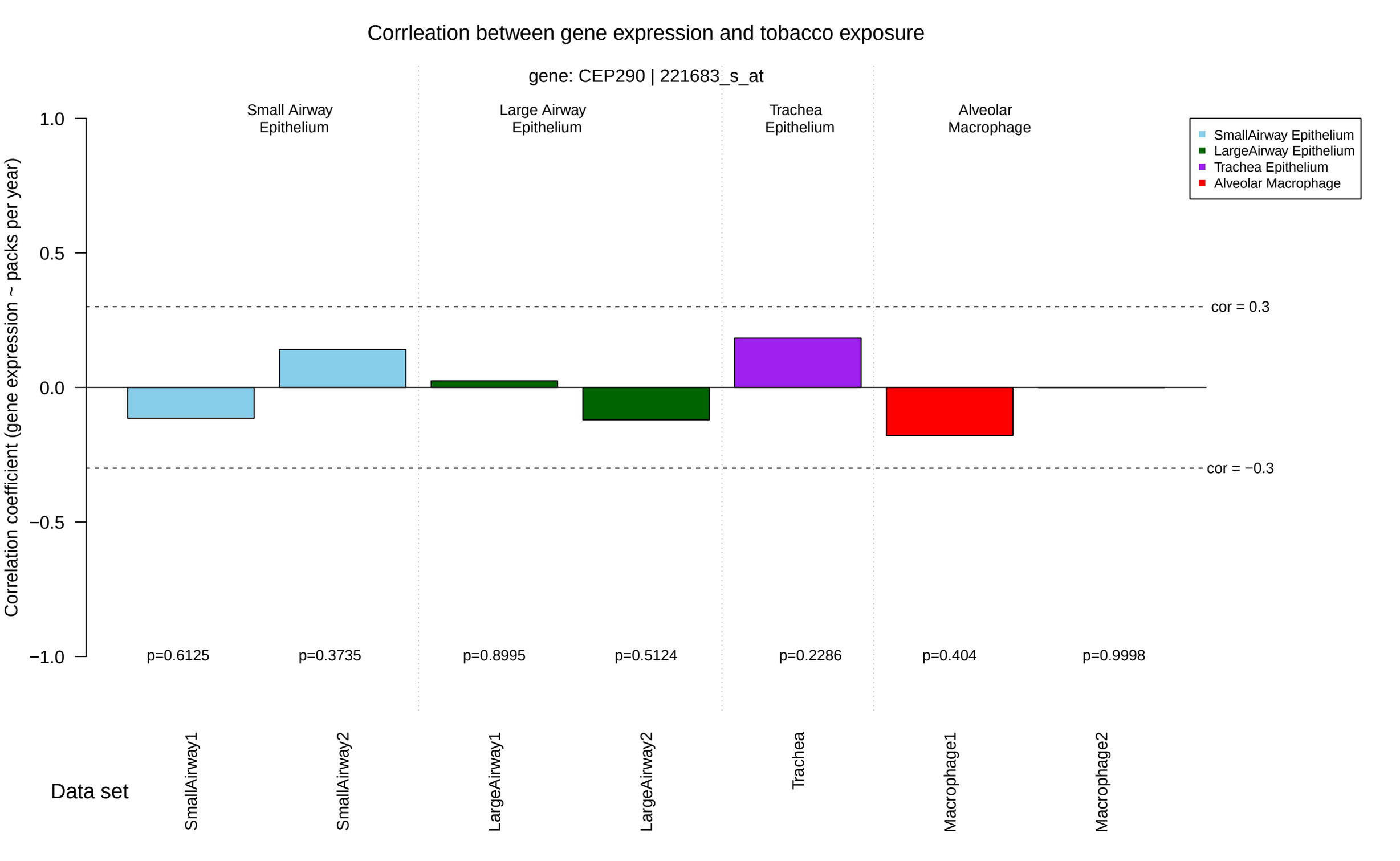

set: LargeAirway1 (p=0.3458)
CEP290 | 221683_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3058)
CEP290 | 221683_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1095)
CEP290 | 221683_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.311)
CEP290 | 221683_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4085)
CEP290 | 221683_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEP290 | 221683_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6125
p=0.3735
p=0.8995
p=0.5124
p=0.2286
p=0.404
p=0.9998
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2